

FRESHWAY

Performance Metrics, Statistical Validation, Significance Testing, and Ablation Analysis

EXPERIMENTAL EVALUATION & STATISTICAL REPORT

Quantitative Model Metrics, ANOVA/t-test Validation, and Component Ablation

Data-Driven Validation Companion Document

Performance Testing & Model Generalization Analysis

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FreshWay: Empirical Performance, Statistical Analysis, and Experimental Evaluation

This report presents a rigorous evaluation of the FreshWay hybrid quality assessment system. We analyze the object detection and classification models, calculate standard metrics (accuracy, precision, recall, F1), present confusion matrices and ROC curves, document cross-validation results, perform statistical significance testing, and conduct an ablation study of the pipeline components.

Section 1: Experimental Evaluation Setup

Hardware Configuration

Experiments were executed on a CPU-based evaluation environment to simulate server-side processing conditions for edge deployment:

- **Processor**: Intel Core i7-10700K CPU @ 3.80GHz (8 Cores, 16 Threads)
- **RAM**: 16 GB DDR4
- **Operating System**: Windows 10 Professional
- **Inference Mode**: Single-thread sequential CPU execution (no GPU acceleration, mirroring low-cost cloud hosts)

Dataset Overview

The dataset contains a balanced set of fish eye images representing three target classes: Highly Fresh, Fresh, and Not Fresh.

- **Total Dataset Size**: 3,918 images.
- **Training Set (80%)**: 3,132 images (1,044 images per class).
- **Validation Set (20%)**: 786 images (262 images per class).

Section 2: Model Performance Metrics

We evaluate the performance of both the **YOLOv8-nano Eye Detector** and the **MobileNetV2 Hybrid Classifier** on the validation set.

1. YOLOv8-nano Eye Detector

The eye detector isolates the Region of Interest (ROI) before classification. It is evaluated using Intersection over Union (IoU) thresholds:

- **mAP@50** (Mean Average Precision at IoU = 0.5): 95.4%
- **mAP@50-95** (Mean Average Precision over IoU range [0.5, 0.95]): 78.2%
- **Detection Precision**: 95.8%
- **Detection Recall**: 94.2%
- **Average Localization Latency**: 82 ms per image.

2. Baseline MobileNetV2 Classifier (CNN-Only)

Running the baseline MobileNetV2 classifier on the validation set without the Multi-View crop ensemble or the OpenCV Expert Rules yields:

- **Overall Baseline Accuracy**: 61.45% (483 out of 786 images correctly classified).
- **Class-wise Baseline Evaluation Summary**:

Target Class	Precision	Recall	F1-Score	Validation Count
Fresh	55.60%	53.05%	54.30%	262
Highly Fresh	67.26%	72.14%	69.61%	262
Not Fresh	60.78%	59.16%	59.96%	262

3. Proposed FreshWay Hybrid Classifier (ITA + Expert Rules)

Activating the Multi-View ITA Crop Ensemble and the OpenCV Expert Rules Engine yields:

- **Overall Hybrid Accuracy**: **92.36%** (726 out of 786 images correctly classified).
- **Class-wise Hybrid Evaluation Summary**:

Target Class	Precision	Recall	F1-Score	Validation Count
Highly Fresh	95.31%	93.13%	94.21%	262
Fresh	86.28%	91.22%	88.68%	262
Not Fresh	95.65%	92.37%	93.98%	262

Section 3: Confusion Matrices

1. Baseline MobileNetV2 (CNN-Only) Confusion Matrix

The rows represent ground-truth classes and the columns represent predictions:

Actual \ Predicted	Fresh	Highly Fresh	Not Fresh	Total Row Count
Fresh	139	57	66	262
Highly Fresh	39	189	34	262
Not Fresh	72	35	155	262

2. Proposed FreshWay Hybrid (ITA + Expert Rules) Confusion Matrix

The rows represent ground-truth classes and the columns represent predictions:

Actual \ Predicted	Highly Fresh	Fresh	Not Fresh	Total Row Count
Highly Fresh	244	18	0	262
Fresh	12	239	11	262
Not Fresh	0	20	242	262

Key Observations:

1. ****CNN Limitations****: The baseline CNN confuses "Not Fresh" and "Fresh" frequently (e.g., 72 actual Not Fresh predicted as Fresh) because glossy glares on dry eyes mimic fresh reflections.
2. ****Hybrid Solution****: The hybrid model completely eliminates extreme errors (\$0\$ Highly Fresh predicted as Not Fresh, and vice versa) through physical veto checks.
3. ****Accuracy Gain****: By using OpenCV to veto misleading visual artifacts, we reduce classification errors from \$303\$ cases in the baseline to only \$60\$ cases in the final model.

Section 4: ROC Curve & Area Under Curve (AUC)

The Receiver Operating Characteristic (ROC) curve evaluates True Positive vs. False Positive Rates. We perform a One-vs-Rest (OvR) analysis to compare AUC values:

Area Under the Curve (AUC) Summary:

Class Quality	Baseline CNN AUC	Proposed Hybrid AUC
Fresh	0.7483	0.9450
Highly Fresh	0.8537	0.9820
Not Fresh	0.8085	0.9760

The hybrid model achieves an average AUC of 0.967, proving its superior discriminative power compared to the baseline (average AUC of 0.803).

Section 5: Statistical Analysis & Standard Deviation

We compute the mean and standard deviation (SD) for the key OpenCV physical parameters across the validation set (262 samples per class).

Physical Metric	Highly Fresh Mean (SD)	Fresh Mean (SD)	Not Fresh Mean (SD)
HSV Saturation (0-255)	128.4 (12.3)	92.1 (14.5)	18.4 (6.2)
Specular Highlight (pixels)	112.0 (25.0)	45.0 (15.0)	420.0 (95.0)
Laplacian Variance	142.1 (18.5)	98.4 (12.4)	19.3 (5.2)
Cornea Opacity Ratio	0.02 (0.01)	0.08 (0.03)	0.82 (0.11)

Section 6: 5-Fold Cross-Validation

We perform 5-fold cross-validation on the dataset to verify generalization:

- **Fold 1 Accuracy**: 91.83%
- **Fold 2 Accuracy**: 92.47%
- **Fold 3 Accuracy**: 93.11%
- **Fold 4 Accuracy**: 91.58%
- **Fold 5 Accuracy**: 92.73%
- **Mean Cross-Validation Accuracy**: **92.34%**
- **Standard Deviation (σ)**: **0.61%**

Section 7: Statistical Significance Testing

We compare the classification accuracy of the **Baseline MobileNetV2 Model** against our **FreshWay Hybrid Model** across the 5 cross-validation folds using a paired t-test.

Empirical Fold Accuracies:

Cross-Validation Fold	Baseline MobileNetV2 (%)	FreshWay Hybrid Model (%)	Accuracy Difference (%)
Fold 1	60.50%	91.83%	31.33%
Fold 2	62.10%	92.47%	30.37%

Fold 3	61.80%	93.11%	31.31%
Fold 4	59.90%	91.58%	31.68%
Fold 5	62.90%	92.73%	29.83%
Mean (SD)	**61.44% (1.21%)**	**92.34% (0.61%)**	**30.90% (0.75%)**

T-Test Results:

- ****Calculated t-statistic****: $t(4) = 91.68$
- ****p-value****: 4.87×10^{-7}

Since the p-value ($p < 0.0001$) is extremely low, we reject the null hypothesis. The improvement in quality assessment accuracy from our hybrid pipeline is ****exceptionally statistically significant****.

Section 8: Ablation Study

We analyze the impact of the ****Multi-View Ensemble (ITA)**** and the ****OpenCV Expert Rules Engine**** on overall classification accuracy.

Configuration ID	Base CNN	Multi-View ITA	OpenCV Expert Rule	Validation Accuracy	Accuracy Gain
A (Baseline)	MobileNetV2	No	No	61.45%	Reference
B (Ensemble)	MobileNetV2	Yes	No	72.10%	+ 10.65%
C (Expert Rules)	MobileNetV2	No	Yes	84.50%	+ 23.05%
D (FreshWay Final)	MobileNetV2	Yes	Yes	**92.36%**	**+ 30.91%**

Key Insights from the Ablation Study:

1. ****Multi-View ITA Impact****: Enabling the Multi-View ITA crop ensemble improves accuracy by ****10.65%****. This proves that averaging predictions over standard, tight, wide, and shifted crops corrects localization offsets.
2. ****OpenCV Expert Rules Impact****: Integrating the physical rules engine yields a ****23.05%**** improvement. This confirms the critical role of physical vetoes (saturation and specularity) in overriding misleading deep learning predictions.
3. ****Synergy****: Activating both modules yields a validation accuracy of ****92.36%**** (a total improvement of ****30.91%**** over the baseline CNN).